CSE 7850: Final Project (Group 4)
MedBench, a Benchmark Dataset for Medical Large Language Models

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Abstract

There exist several Large Language Models (LLMs) that can be utilized for several tasks in the medical research space. Tasks include Natural Language Inference (NLI), Question Answering (QA), Sequence Classification, and many others involving medical data. A very common research idea is to adapt LLMs trained on a particular domain to a new domain and dataset. In this paper we introduce `MedBench`, a benchmark dataset we curated involving multiple sequence classification, question answer, and NLI tasks, all on medical data and tasks. We specifically combine the MedNLI, MedQA, Disease Symptoms, PubMedQA, and BioASQ datasets and additionally use text-based data augmentation to increase the number of samples. We show the performance of three state-of-the-art models—BioGPT, Clinial BERT, and BioBERT—on `MedBench`, showing that they all indicate high performance in these tasks, demonstrating that our dataset can be used to evaluate new LLM architectures designed for medical research.

1. Introduction

Large language models have become an integral part of machine learning advancement, and increasingly so in the medical research space. LLMs have been applied in numerous medical spaces, such as in protein sequence prediction, disease diagnosis, electronic health records, and several other types of data and tasks. Recently, the complexity and adaptability of medical LLMs has propelled the exploration of zero-shot learning and fine-tuning to apply pre-trained models to a completely different task, usually out-of-domain [7]. There are important applications of this in the medical and pharmaceutic space, such as AI-powered doctor chatbots, which would be able to answer any questions a patient has about prescriptions, symptoms, or be able to diagnose a patient based on their symptoms.

Given the fact that LLMs are inherently able to quickly use transfer learning to excel at an out-of-domain task, we aim to identify a cohort of data to test this adaptability for any custom model trained on any type of medical data. The number of existing benchmark datasets is quite limited and often doesn't include certain types of tasks LLMs are suited to, which include natural language inference (NLI), question-answering (QA), and text-based classification.

By combining a variety of tasks into a single benchmarking dataset we call `MedBench`, we aim to move beyond the traditional, siloed approach of model evaluation and create a unified solution for all LLMs to be tested for adaptability. This holistic testing mirrors the multifaceted challenges faced by medical professionals in real life, where understanding patient records, making diagnostic decisions, and answering questions are a significant part of their daily routines. Moreover, this benchmarking dataset will provide a more comprehensive view of each model's capabilities.

2. Literature Review

While there exist benchmark datasets for medical Large Language Models, our literature review only found datasets that either tested a specific type of natural language task or had similar types of data. One example of a popular benchmarking dataset for medical LLMs is the OpenMedical-LLM [11], a leaderboard created collaboratively by researchers at Open Life Science AI, University of Edinburgh, and HuggingFace. This leaderboard utilizes datasets from a wide array of sources, including MedQA, PubMedQA, and other datasets specifically for question answering. This is quite a large and substantial dataset including topics such as biochemistry, pediatrics, physics, and many large language models such as Google's Gemini Pro have been properly evaluated in these domains. However, a key potential improvement we identify in OpenMedical-LLM is the addition of new types of tasks (i.e more than question answering).

A similar dataset, MultiMedQA, was created a group of researchers at Google to test their model Med-PaLM [13]. The researchers use many of the datasets used in OpenMedical-LLM, including MedQA and PubMedQA, alongside the MMLU dataset, which contains multiple choice questions from general topics including genetics and college biology courses. While this was demonstrated to test Med-PaLM and other models quite well, it still would not test models' adaptability to tasks such as disease prognosis classification or medical natural language inference.

However, given that these two performant studies both utilize the MedQA and PubMedQA datasets, we decided to closely examine these two datasets. Di Jin et. al, a group of researchers from MIT, created MedQA as an open-domain QA dataset, specifically from medical exams [5]. MedQA is a multilingual dataset containing questions written in English, traditional Chinese, and simplified Chinese. Each entry in MedQA includes the following parts: the question, answer candidates (as a multiple choice), and a document containing sources of text that are relevant to the question. PubMedQA was developed by Qiao Jin et. al [6], and is a dataset containing QA prompts from abstracts of articles on PubMed, a database of biomedical research literature. The authors specifically utilize articles on PubMed that have a question in their abstract along with language that allows for the formation of a question.

Original Statement Title	Converted Question	Label	%
Spontaneous electrocardiogram alterations ***predict*** ventricular fibrillation in Brugada syndrome.	***Do*** spontaneous electrocardiogram alterations ***predict*** ventricular fibrillation in Brugada syndrome?	***yes***	92.8
Liver grafts from selected older donors ***do not have*** significantly more ischaemia reperfusion injury.	***Do*** liver grafts from selected older donors ***have*** significantly more ischaemia reperfusion injury?	***no***	7.2

Figure 1: Example of a statement in a PubMed abstract to generated question in PubMedQA.

3. Dataset Components

Utilizing our findings from the literature review, we decided to start our benchmark dataset by including both PubMedQA and MedQA, as they are the state-of-the-art in question-answering datasets for medical purposes. We also decided to utilize the Bio-ASQ dataset, which is a large-scale crowdfunded biomedical question-answering dataset supervised by Tsatsaronis et. al [14]. It is essentially an extension of the PubMedQA, but using a large number of participants to carefully tune the dataset using several heuristic factors such as mean average precision and the ROUGE score, which compares a generated text-summary against a list of references [9]. For all these QA tasks, we select the yes/no answer column instead of the long answer column for training, and include the context alongside the query in the prompt.

The next dataset we identified was MedNLI, a dataset for medical natural language inference created by Romanv and Shivade [12]. The dataset was annotated by doctors, and each entry in the dataset contains a sentence containing the premise, a sentence containing the hypothesis, and one with a label indicating whether the hypothesis entails, contradicts, or is not affected (neutral) by the premise. Released in 2019, this dataset is now one of the most popular datasets used to test natural language inference tasks on medical LLMs.

Given that we curated datasets for question answering and natural language, the final dataset we identified was for a multi-classification task. We found a popular dataset on GitHub for disease prognosis created by Anuj Dutt [2], which we'll refer to as "Disease Symptoms" going forward in this paper. It has over 250 stars from GitHub users. The training and testing data CSV has 133 columns, 132 of which are different symptoms, and the last column is the prognosis of over 41 different types of diseases. Each row is a one-hot encoded vector of which synonyms the patient experienced. The data was originally sourced from a Kaggle dataset containing a subset of the data as well as a collection of records from the New York Presbyterian Hospital curated by a researcher at Columbia University.

To summarize, our `MedBench` dataset consists of PubMedQA, MedQA, BioASQ, MedNLI, and Disease Symptoms. It covers a wide variety of space in the medical field, from entrance exams to hospital records, as well as a multitude of natural language tasks. We aim to have this dataset be a good indicator of how versatile and performant a LLM is on several types of tasks.

4. Methods and Results

4.1. Data Augmentation

We decided to apply data augmentation and modification to the original datasets that we curated, to both allow the

sentence	prognosis
The patient has skin rash, pus filled pimples, blackheads, and scurring.	Acne
The patient has muscle weakness, stiff neck, swelling joints, movement stiffness, and painful walking.	Arthritis
The patient has burning micturition, bladder discomfort, and continuous feel of urine.	Urinary tract infection
The patient has vomiting, sunken eyes, dehydration, and diarrhoea.	Gastroenteritis
The patient has chills, vomiting, fatigue, weight loss, cough, high fever, breathlessness, sweating, loss of appetite, mild fever, yellowing of eyes, swelled lymph nodes, malaise, phlegm, chest pain, and blood in sputum.	Tuberculosis
The patient has skin rash, pus filled pimples, and blackheads.	Acne
The patient has acidity, indigestion, headache, blurred and distorted vision, excessive hunger, stiff neck, depression, irritability, and visual disturbances.	Migraine
The patient has chills, fatigue, high fever, headache, nausea, constipation, abdominal pain, diarrhoea, toxic look (typhos), and belly pain.	Typhoid
The patient has joint pain, vomiting, yellowish skin, dark urine, nausea, loss of appetite, abdominal pain, diarrhoea, mild fever, yellowing of eyes, and muscle pain.	hepatitis A
The patient has skin rash, joint pain, skin peeling, silver like dusting, small dents in nails, and inflammatory nails.	Psoriasis
The patient has vomiting, headache, and weakness of one body side.	Paralysis (brain hemorrhage)
The patient has indigestion, loss of appetite, abdominal pain, passage of gases, and internal itching.	Peptic ulcer diseae

Figure 2: Snippet of the modified Disease Symptoms dataset with sentence construction.

large language models we selected in the next section to easily handle the data, as well as add more variability to the dataset. First, we utilized sentence negation as a form of text augmentation for both the BioASQ and PubMedQA datasets. Text negation has been demonstrated to be a form of text augmentation that increases complexity and model performance [3]. We duplicated each entry in the BioASQ and PubMedQA datasets by applying a negation function provided by the Python negate package, which internally uses both heuristics and a encoder-decoder transformer architecture to generate a sentence negated from the input.

Additionally, to convert the Disease Symptoms dataset into a prompt for the LLM, we convert each one-hot vector row in the dataset into a sentence with the following format: "The patient has *symptom1*, *symptom2*, . . . " where the symptoms correspond to the active columns in the vector. Then, it becomes a multiclassification task from a prompt, which is more easily adapted to a natural language task then before. Lastly, for each of these datasets, we compiled both a train and test dataset.

4.2. Model Selection

We selected three large language models to test with our benchmark. The first was BioGPT, a large language model developed by a team of researchers at Microsoft [10] that utilizes the generate pre-trained transformer architecture introduced by OpenAI. BioGPT was trained exclusively on biomedical literature, and so should be very well suited to the series of tasks we include in `MedBench`. BioGPT was trained on a variety of datasets and tasks, and even achieved a record of 78.2% accuracy on PubMedQA. The researchers specifically implemented BioGPT to be adaptable to unseen down-stream tasks by introducing a framework which is shown below.

The second model we identified was ClinicalBERT, a model developed by Huang, Altosaar, and Ranganath [4] which is based on the "Bidirectional Encoder Representations from Transformers" model architecture introduced by Google [1]. The researchers pretrain this model on a set of clinical notes, and then fine-tune it for the downstream task of predicting patient readmission into hospitals. Given that it was trained on clinical data, our aim was to see its efficacy and performance on different types of data such as general biomedical knowledge present in the PubMedQA dataset.

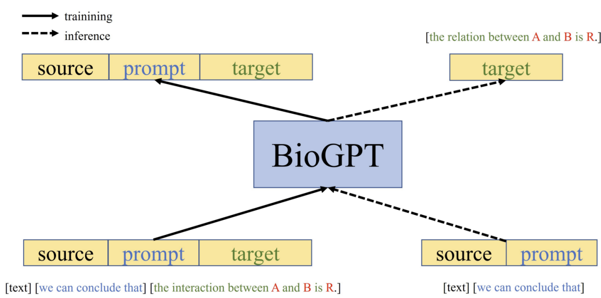


Figure 3: Image from the BioGPT paper that depicts the downstream adaptation framework.

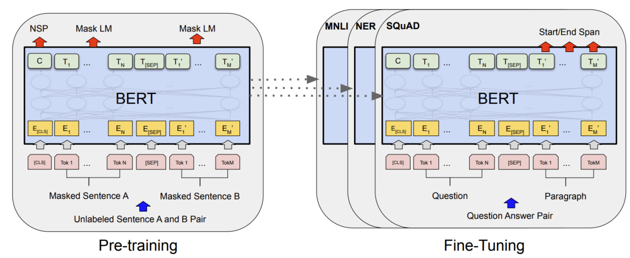


Figure 4: BERT architecture and pretraining/fine-tuning methods. Visual from original BERT paper.

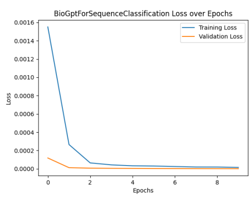
The final model we selected was BioBERT, another BERT-architecture model developed by Lee et. al [8]. They pretrained this model on PubMed abstracts as well as full-text articles on PubMed. The model was then pretrained on named entity recognition, relation extraction, and question answering tasks. Given that this model is the most tuned to understanding complex biological terms (as it was trained on research article), we expect it to be quite resilient to the different types of datasets in `MedBench`.

4.3. Training and Results

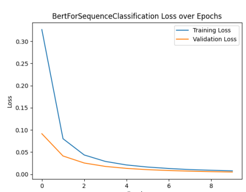
We utilized Huggingface's transformers library to import the pretrained models for BioGPT, ClinicalBERT, and BioBERT. For each of the individual datasets in `MedBench`, we fine-tune each of the three models, all using the same hyperparameters. Each of the training datasets was split into a train and valid dataset using scikit-learn's train_test_split function with a split size of 0.2.

4.3.1 Disease Symptoms

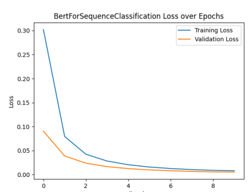
Using a node with a single NVIDIA Tesla V100 GPU with 16GB VRAM on the Georgia Tech PACE cluster, we trained all three models for 10 epochs on the full training dataset. As the dataset's individual prompts were relatively small, we were able utilize a training batch size of 32 without running into any GPU memory issues. We utilized a learning rate of $1\mathrm{e}{-4}$, the Adam optimizer, and a categorical cross entropy loss function. We utilized accuracy for a more visually intuitive method of examining model performance. The training curves and test set evaluation results are shown below. All models performed incredibly well, reaching 100% validation accuracy after just the second epoch of training and reaching 100% accuracy on the test dataset.



(a) BioGPT



(b) BioBERT



(c) ClinicalBERT

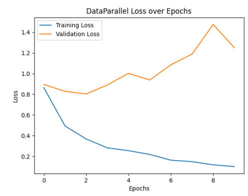
Figure 5: Training curves on Disease Symptoms dataset.

Table 1: Disease Symptoms test set evaluation

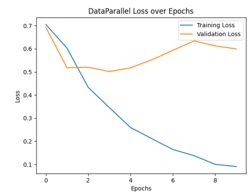
Model	Test Loss	Test Accuracy
BioGPT	$1.816\mathrm{e}{-6}$	100%
ClinicalBERT	$4.979\mathrm{e}{-3}$	100%
BioBERT	$5.156\mathrm{e}{-3}$	100%

4.3.2 BioASQ

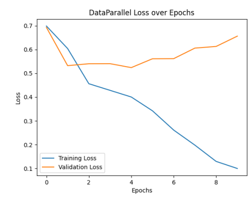
Similarly, we trained all three models on the BioASQ dataset. Due to the prescence of long contexts, we utilized 2 NVIDIA RTX 6000 GPUs with 24GB VRAM each on the PACE cluster. For each of the three models, we had to limit the tokenized length of the prompt to 500 characters to avoid running out of GPU memory, which did result in some inaccuracy and overfitting in the model (as some of the context was cut out of the prompt). We utilized a batch size of 16, learning rate of $1\mathrm{e}{-5}$, 10 epochs, the Adam optimizer, and the binary cross-entropy loss. Similarly, we utilized accuracy to more intuitively interpret model performance. All three models began overfitting very easily, but still resulted in good test accuracies for question-answering.



(a) BioGPT



(b) BioBERT



(c) ClinicalBERT

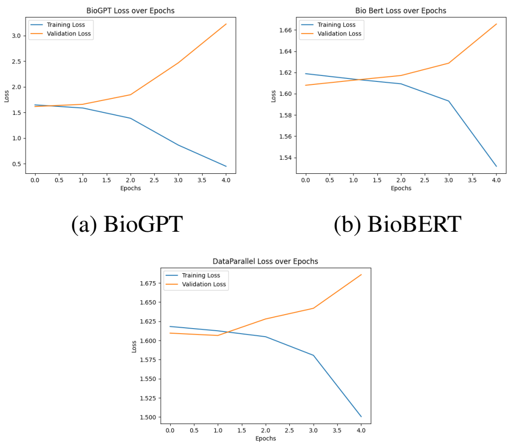
Figure 6: Training curves on BioASQ dataset.

Table 2: BioASQ test set evaluation

Model	Test Loss	Test Accuracy
BioGPT	0.9296	71.64%
ClinicalBERT	0.6009	82.34%
BioBERT	0.5729	84.08%

4.3.3 MedQA

Next, we trained all three models on the MedQA dataset, which consisted of multiple choice questions from medical licensing exams, like the USMLE Step 1, 2, and 3. Since the questions had a lot of background detail included as well as the five answer choices appended, we utilized 4 NVIDIA Quadro RTX 6000 GPUs with 24GB VRAM from the PACE cluster. For the BERT-based models, we had to limit the tokenized length of the question to 512 characters to avoid running out of GPU memory. We utilized a training batch size of 16 (larger batch sizes resulted in OutOfMemory errors), learning rate of $1\mathrm{e}{-5}$, 5 epochs, the Adam optimizer, and cross-entropy loss. We kept track of training and validation loss and accuracy to gauge model performance. The validation loss increased overtime while the training loss decreased, showing clear signs of overfitting for all three models. As a result, all the models failed to perform well in this task.

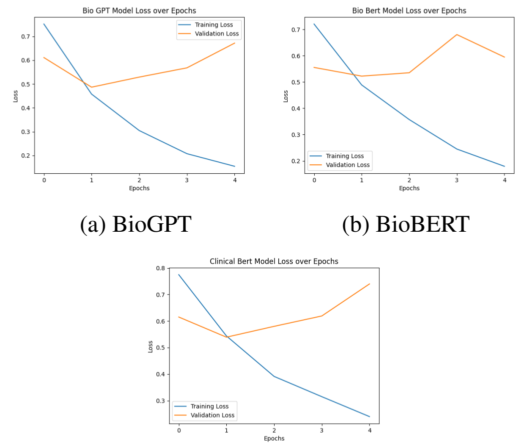


(a) BioGPT

(b) BioBERT

(c) ClinicalBERT

Figure 7: Training curves on MedQA dataset.



(a) BioGPT

(b) BioBERT

(c) ClinicalBERT

Figure 8: Training curves on MedNLI dataset.

Table 3: MedQA test set evaluation

Model	Test Loss	Test Accuracy
BioGPT	3.1718	20.58%
ClinicalBERT	1.7159	18.22%
BioBERT	1.6992	19.95%

4.3.4 MedNLI

Once again, we trained all three models on the MedNLI dataset. This dataset is composed of three things: premise, hypothesis, and labels. To train in an effective manner, we utilized 4 NVIDIA Quadro RTX 6000 GPUSs with 24GB VRAM from the PACE cluster. There were originally many issues with GPU out of memory error, but these errors were eventually resolved when we had decreased to training batch size to 32, validation batch size to 256, and testing batch size to 256. Furthermore, the models were all trained with a learning rate of $1\mathrm{e}{-5}$, 5 epochs, the Adam Optimizer, and a cross entropy loss function. When looking at the graph shown below, we can see that towards the end of the last epoch, the model had begun to overfit as the training loss was decreasing but the validation loss was slightly increasing. However, the training and validation accuracies of the model increase, and thus allowed the model to perform well on the MedNLI dataset.

Table 4: MedQA test set evaluation

Model	Test Loss	Test Accuracy
BioGPT	0.7415	79.54%
ClinicalBERT	0.7698	77.92%
BioBERT	0.5855	80.52%

4.3.5 PubMedQA

We were unable to train the three models on the PubMedQA dataset due to its sheer size. Post augmentation, the dataset consisted of over 400,000 data-points, and each yes/no question was paired with an extensive context dictionary. Utilizing 4 NVIDIA Quadro RTX 6000 GPUs with 24 GB of VRAM on the PACE cluster, we continuously ran into excess memory allocation errors. In order to resolve these errors, we tried various strategies, including selecting just the most significant field from the context dictionary, tokenizing a minimal amount of context information, and decreasing batch size. However, we were unsuccessful in our ability to train and test the three models on PubMedQA because of the limitation on our compute resources. As a result, MedBench consists of data sourced from four datasets: Disease Symptoms, BioASQ, MedQA, and MedNLI.

5. Discussion and Conclusion

Overall, after training and testing the models on the datasets we curated, we see that the model had performed well for the datasets, Disease Symptoms, BioASQ, and MedNLI. This indicates that there exists a significant alignment between model training and task-specific requirements. Furthermore, we see that popular models such as BioGPT, ClinicalBERT, and BioBERT all perform well when tested on these datasets.

However, the performance on the MedQA dataset was significantly lower than that of the other datasets. We see that the test accuracy was around 20 %. Since there are five answer options in the dataset, the model's accuracy is very similar to the accuracy of random guessing. The reason for this is that datasets like Disease Symptoms, BioASQ, and MedNLI all had constant options. For example, questions in BioASQ had a yes or no answer and questions in the MedNLI had a entailment, contradiction, and neutral answer choices. On the other hand, questions in the MedQA dataset had five different answer options that were specific to each question. Thus, MedQA creates an increased challenge level for models for evaluation that may need further research if they were to be included in the benchmarking dataset. MedBench will now only include Disease Symptoms, BioASQ, and MedNLI, making this benchmarking dataset an effective way to validate the performance of biology-related large language models.

6. Contributions

Here are the individual contributions for each team member in the project:

- Saigautam: Trained all three selected models on both the disease symptoms and bioASQ datasets using two parallelized GPUs on the PACE cluster. Lead on the paper writing.

- Adityaa: Sourced the MedQA dataset. Trained all three selected models on the MedQA dataset using parallelization across four GPUs from the PACE cluster.

- Advik: Trained all three selected models on the MedNLI dataset using parallelization across four GPUs from the PACE cluster.

- Mihir: Sourced the pre-trained models and datasets for MedBench. Preprocessed the curated data. Attempted to train all three selected models on PubMedQA using parallelization across four GPUs on the PACE cluster.

7. Source Code

The dataset and training code we used can be accessed via the following GitHub repository link: `https://github.com/saisree27/med-bench`.

8. Differences from Project Proposal

We changed a couple ideas and methods from our original idea submitted in the project proposal:

- The datasets we identified in the proposal included PubMedQA, BioASQ, OpenI, and MedNLI. We ultimately changed this by removing the OpenI dataset and adding the MedQA and Disease Symptom datasets.

- In our proposal we identified different loss metrics for each subdataset of the benchmark dataset we developed. Instead of doing this, we decided to standardize this to categorical cross-entropy loss and accuracy for better interpretability across tasks.

- Our initial goal was to utilize Google Colab for all our training, but we quickly realized that the dataset and the size of the transformer models forced Colab's free tier to run out of memory easily. As a result, we pivoted to using the GPUs in the PACE cluster, which was a new experience for many of our group members involved a learning curve. However, now we are much more familiar with the PACE cluster and how to effectively run ML training on it.

References

[1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019.

[2] Anuj Dutt and Shivani P. Disease prediction from symptoms. `https://github.com/anujdutt9/Disease-Prediction-from-Symptoms`, 2023.

[3] Giannis Haralabopoulos, Mercedes Torres Torres, Ioannis Anagnostopoulos, and Derek McAuley. Text data augmentations: Permutation, antonyms and negation. *Expert Systems with Applications*, 177:114769, 2021.

[4] Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. Clinicalbert: Modeling clinical notes and predicting hospital readmission, 2020.

[5] Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams, 2020.

[6] Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William W. Cohen, and Xinghua Lu. Pubmedqa: A dataset for biomedical research question answering, 2019.

[7] Mert Karabacak and Konstantinos Margetis. Embracing large language models for medical applications: Opportunities and challenges. *Cureus*, 15(5):e39305, May 2023.

[8] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, Sept. 2019.

[9] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics.

[10] Renqian Luo, Liai Sun, Yingce Xia, Tao Qin, Sheng Zhang, Hoifung Poon, and Tie-Yan Liu. BioGPT: generative pre-trained transformer for biomedical text generation and mining. *Briefings in Bioinformatics*, 23(6), 09 2022. bbac409.

[11] Ankit Pal, Pasquale Minervini, Andreas Geert Motzfeldt, Aryo Pradipta Gema, and Beatrice Alex. openlifescienceai/open_medical_llm_leaderboard. https://huggingface.co/spaces/openlifescienceai/open_medical_llm_leaderboard, 2024.

[12] Alexey Romanov and Chaitanya Shivade. Lessons from natural language inference in the clinical domain.

[13] Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, Perry Payne, Martin Seneviratne, Paul Gamble, Chris Kelly, Abubakr Babiker, Nathanael Schärli, Aakanksha Chowdhery, Philip Mansfield, Dina Demner-Fushman, Blaise Agüera Y Arcas, Dale Webster, Greg S Corrado, Yossi Matias, Katherine Chou, Juraj Gottweis, Nenad Tomasev, Yun Liu, Alvin Rajkomar, Joelle Barral, Christopher Semturs, Alan Karthikesalingam, and Vivek Natarajan. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180, Aug. 2023.

[14] George Tsatsaronis, Georgios Balikas, Prodromos Malakasiotis, Ioannis Partalas, Matthias Zschunke, Michael R Alvers, Dirk Weissenborn, Anastasia Krithara, Sergios Petridis, Dimitris Polychronopoulos, Yannis Almirantis, John Pavlopoulos, Nicolas Baskiotis, Patrick Gallinari, Thierry Artiéres, Axel-Cyrille Ngonga Ngomo, Norman Heino, Eric Gaussier, Liliana Barrio-Alvers, Michael Schroeder, Ion Androutsopoulos, and Georgios Paliouras. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. *BMC Bioinformatics*, 16(1):138, Apr. 2015.